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Prepared by: Prof. Arjun Chavada, DU

Unit – 4A ➡ Water Pollution

Introduction

Introduction:

- Environmental pollution means “the addition of any substance or any form of energy into the environment at a rate faster than it can be dispersed/decomposed in a harmless form.”
- Any change which adversely affects living organisms is known as pollution.
- Pollution may be natural (due to natural calamity) or man-made (human activity).
- Substance that causes pollution is known as a “**pollutant**”.
- Not all pollutants are harmful, if not present in excess concentration.

Types of pollution:

- If any of the components of the environment gets affected, subsequently living organisms are also affected, thus pollution occurs.
- As discussed, pollution can be natural (volcanic eruption, UV radiation, forest fire) or man-made.
- In the beginning, nature was able to absorb harmful substances, but after industrialization, overpopulation, and urbanization nature was unable to process or neutralize harmful by-products of human activity.
- Pollution in various components can be classified as;

(1) Water pollution

(2) Air pollution

(3) Land pollution

(4) Noise pollution

(5) Thermal pollution

Water Pollution

- Water is one of the necessities of survival.
- Water is consumed in every sector to produce goods and various services.
- Water used for drinking by living organisms is known as “**Potable Water**”.
- Alteration in the physical, chemical, or biological characteristics of water through natural or man-made activity is known as “**water pollution**”.
- Presence of harmful substances in water causing diseases/ill-effects to a living organism is known as “**water pollution**”.
- Addition of chemicals decreases the quality of the water, causing pollution. These pollutants can be of various forms.

Water quality standards

- Fresh water on earth is put to many uses such as drinking, bathing, cooking, agriculture, and industrial purposes.
- The used water becomes contaminated and is called wastewater.
- Water quality depends upon the proposed use of the water, whether for drinking or industrial use, irrigation, etc.
- Characteristics of potable water are;
 - (1) It should be colorless.
 - (2) It should be odorless.
 - (3) It should be free from suspended solids.
 - (4) It should have good taste.
 - (5) It should not contain any harmful substances or pathogens.

Water quality parameters

- Main 3 types of parameters are required to be tested to assess the quality of water

(1) Physical

- Turbidity
- Colour
- Odour
- Taste

- Temperature

(2) Chemical

- Solids (suspended, dissolved, colloidal)
- Hardness
- Chloride
- pH
- Other chemical substances (nitrogen, fluoride, metals)

(3) Biological

- Various micro-organisms (bacteria, fungi, viruses)
- Pathogen (disease-causing bacteria, viruses)

Sources of Water Pollution

→ Sources of water pollution can be classified as;

(1) Point sources:

→ Those sources identified as a single location are called “point sources”.

→ For example;

- Industrial effluents
- Power plant
- Sewage discharge

→ It is possible to minimize water pollution from point sources if the wastewater is collected and appropriately treated before discharging into the water bodies.

(2) Non-point sources:

→ The discharge from these sources is not at any particular site, somewhat of these sources are scattered, which individually or collectively pollute the water.

→ For example;

- Surface run-off from agricultural field
- Overflowing small drains
- Rainwater sweeping roads and fields
- Atmospheric pollutants getting dissolved in water

→ Major sources of water pollution are;

(1) Sewage: discharge from sewer and drains

(2) Industrial effluents from different industries

(3) Synthetic detergents used for washing and cleaning

(4) Agro-chemicals like fertilizers containing nitrates and phosphate and pesticides

(5) Oil spilling during drilling and shipment

(6) Waste heat from industrial discharge increases the temperature of the water bodies

→ Major sources of groundwater pollution are septic tanks, mines, and deep well injection of effluents.

Classification of Water Pollutants

→ Water pollutants are classified into the following categories;

(1) Suspended matter

→ Suspended matter are substances that are freely moving in the region of water.

→ It can be soil particles, rock fragments, parts of vegetation, etc.

(2) Natural organic pollutants

→ Sewage and industry effluent composed of organic substances

(3) Synthetic organic compounds

→ Detergent, fertilizers, pesticides, etc.

(4) Pathogens

→ Disease-causing bacteria, viruses, protozoa, etc.

(5) Inorganic matter

→ Alkali, acid, and non-carbohydrate substances in general.

(6) Thermal pollutants

→ Substances (solid, liquid) due to which the temperature of the water increases.

(7) Radioactive pollutants

→ Matter which emits radiation and is further harmful to water quality

(8) Sediments

→ Small organic and inorganic particles

→ General classification based on physical nature is

- **Suspended:**

- these particles have a diameter of size more than 1 micrometer.
- They are large enough to settle out of the water quickly and may be retained on the filter.

- **Colloidal:**

- These particles are so small that their settlement is insignificant.

- To remove this type of impurity sedimentation with coagulation is adapted.
- Dissolved:
 - These pollutants get dissolved in the water and make the water turbid, such matter does not settle out nor can be removed by filters.
 - It can be organic or inorganic.

Effects of different pollutants:

Organic pollutants:

- As the organic pollutants can be decomposed it is also known as oxygen demanding waste.
- Organic matter which reaches water bodies is decomposed by micro-organisms (bacteria) present in the water.

- **Biochemical Oxygen demand (BOD)** is used to as a measure to find out the amount of Biodegradable matter in the water.



oxygen



Organic matter



Micro-organisms

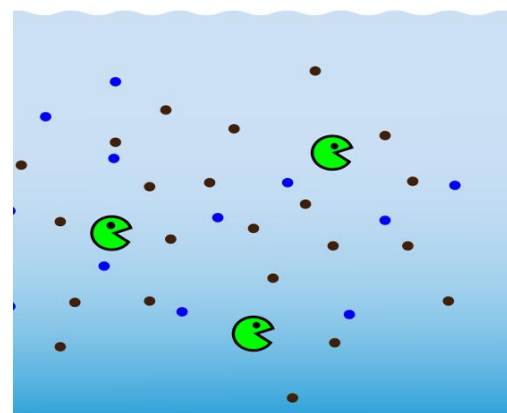
- It is defined as amount of oxygen required by the bacteria to decompose the biodegradable organic matter under aerobic condition.

- The saturated value of DO in water is in the order of 8 to 10 mg/L.

- Optimum DO require for healthy fish and other aquatic life is about 5-8 mg/L.

- If DO drops below 4 mg/L fish and other aquatic life is threatened and in extreme cases killed.

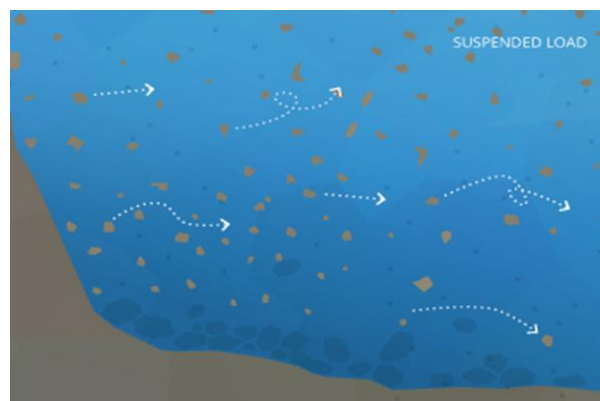
- Other effect of reduced DO is undesirable taste, odour, colour, which prohibits its use for domestic and recreational purpose.



Suspended matter:

- Suspended matter has following effect on the quality of water;

1. It makes water aesthetically displeasing
2. Biodegradable suspended matter causes DO depletion

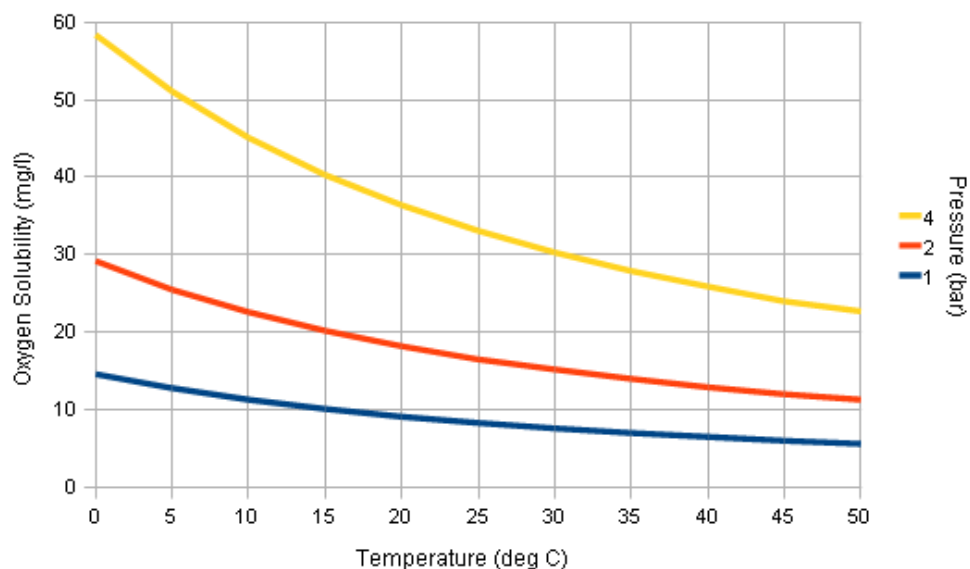


3. It reduces light penetration, affecting photosynthesis
4. Provide absorption sites for biological organism, affecting flora fauna.

Thermal Pollutants:

→ Water is used for dissipation of waste heat in power plants and other industry, which is subsequently discharged into streams causing following ill effects.

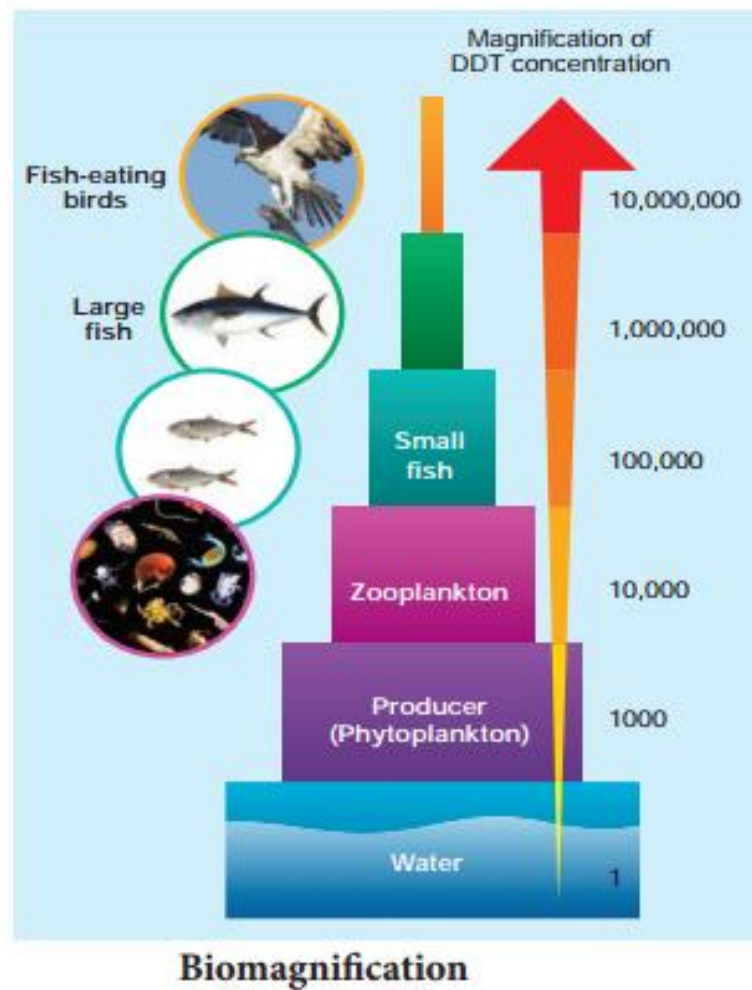
1. Increase in biological activity
2. Death of small organism
3. Saturation DO decrease



4. Growth of algae is accelerated

Toxic Compounds:

- Pollutants such as heavy metals, pesticides, cyanides are harmful to aquatic organisms.
- Substances like pesticides, methyl mercury etc., moves in to the bodies of organisms from medium in which these organisms live.
- These substances tend to accumulate in the organism's body. This process is called "**Bio accumulation**".
- The concentration of these toxic substances builds up at successive levels of food chain. This process is called "**Biomagnification**".
- A disease called "**Minamata**" can be caused by consuming methyl mercury contaminated fish.



Pathogens:

- Many wastes water, especially sewage, contains many pathogenic micro-organisms.
- Presence of such micro-organism causes various water borne diseases like
 1. Cholera
 2. Dysentery
 3. Typhoid
 4. Hepatitis A
 5. Giardia

Eutrophication

Introduction:

- Presence of nutrients is must for growth of organisms, but if these nutrients are present in excessive amount, then they act as pollutants
- they allow excessive growth of aquatic plants like algae.
- The word Eutrophication is originated from Greek words eu=well and trophes= food.
- Thus, meaning is “well fed” or “nutrient rich”.



- So, Eutrophication is **excessive nutrient load** in a water body or enrichment of water body by nutrients like phosphorus and nitrogen.

Stages of eutrophication:

- Depending upon the presence of nutrients, the water bodies (aquatic system) may be in one of the following stages,
 1. Oligotrophic: Water bodies with poor concentration of nutrients and very low productivity of aquatic plants.
 2. Mesotrophic: Water bodies with moderate concentration of nutrients and average productivity of aquatic plants.
 3. Eutrophic: Water bodies with high concentration of nutrients and very high productivity of aquatic plants.
- A young lake or a newly formed lake has low nutrient content and has low plant productivity.
- With the passage of time such lakes acquire nutrients from surface runoff from agricultural field, domestic sewage, etc. which ultimately increase aquatic growth.
- In this way oligotrophic water bodies turn mesotrophic water bodies.

Unit 4A – Water pollution

- Eutrophication is natural process which takes many of years, but this is greatly enhanced by human activities and is called accelerated eutrophication.
- The industrial waste, municipal waste, runoff from agricultural land provide plenty of phosphate, nitrates and various minerals stimulating algal growth. Due to which water becomes green.
- In due course of time the algal boom may die abruptly.
- The decaying of organic matter causes depletion of DO, destroying fish and other aquatic species.

Marine Pollution

- Sea and ocean which covers about two third of the surface of the earth plays a vital role in survival of more than 2,50,000 marine species.
- Man uses sea for different purposes like food, adventure, commerce, industry and recreation.
- It is major sink or reservoir of all the pollutants. Pollutants can enter the sea directly from the outfalls and sometimes from costal town, but most frequently they enter from estuaries.
- Definition – “Introduction of substances to marine environment directly or indirectly by humans, resulting in adverse effect on marine activity and lowering of sea water quality and ecosystem.”
- Marine pollutants can be
 1. Oil
 2. Heavy metal
 3. Heat
 4. Radioactive substance
 5. Organic waste
 6. Pesticides

Control of water pollution

- By proper sewage treatment: sewage should be properly treated before disposing it in any water bodies. Sewage should be given following treatment before discharging into water bodies
 1. Primary treatment: To remove floating impurities, grit, inorganic particles, settleable solids etc. It mainly involves physical method for removal of impurities.
 2. Secondary treatment: Mostly aimed to remove organic impurities using mainly biological methods.
 3. Tertiary treatment: Mostly using strong oxidizing agents to remove impurities. Example chlorine gas. It is used to remove the impurities remaining after primary and secondary treatment.
- The industrial effluent should be properly treated before discharging it into water bodies.
- Increasing the vegetative cover to reduce water pollution due to soil erosion.
- Encouraging reuse of water.
- To avoid thermal pollution water should be cooled before discharging it into the streams.
- Advance treatments to remove nitrates and phosphates to prevent eutrophication of the water bodies.